value group of v. We say that v is a *discrete valuation* if its value group is equal to $\mathbb{Z}$ (every discrete subgroup of $\mathbb{R}$ is isomorphic to $\mathbb{Z}$, so we can always rescale a valuation with a discrete value group so that this holds). Given a field k with valuation v, the set

$$A := \{x \in k : v(x) \geq 0\},$$

is the *valuation ring* of k (with respect to v). A *discrete valuation ring* (DVR) is an integral domain that is the valuation ring of its fraction field with respect to a discrete valuation; such a ring A cannot be a field, since $v(\operatorname{Frac} A) = \mathbb{Z} \neq \mathbb{Z}_{\geq 0} = v(A)$.

It is easy to verify that every valuation ring A is a in fact a ring, and even an integral domain (if x and y are nonzero then $v(xy) = v(x) + v(y) \neq \infty$, so $xy \neq 0$), with k as its fraction field. Notice that for any $x \in k^\times$ we have $v(1/x) = v(1) - v(x) = -v(x)$, so at least one of x and $1/x$ has nonnegative valuation and lies in A. It follows that $x \in A$ is invertible (in A) if and only if $v(x) = 0$, hence the unit group of A is

$$A^\times = \{x \in k : v(x) = 0\},$$

We can partition the nonzero elements of k according to the sign of their valuation. Elements with valuation zero are units in A, elements with positive valuation are non-units in A, and elements with negative valuation do not lie in A, but their multiplicative inverses are non-units in A. This leads to a more general notion of a valuation ring.

Definition 1.11. A *valuation ring* is an integral domain A with fraction field k with the property that for every $x \in k$, either $x \in A$ or $x^{-1} \in A$.

Let us now suppose that the integral domain A is the valuation ring of its fraction field with respect to some discrete valuation v (which we shall see is uniquely determined). Any element $\pi \in A$ for which $v(\pi) = 1$ is called a *uniformizer*. Uniformizers exist, since $v(A) = \mathbb{Z}_{\geq 0}$. If we fix a uniformizer π, every $x \in k^\times$ can be written uniquely as

$$x = u\pi^n$$

where $n = v(x)$ and $u = x/\pi^n \in A^\times$ and uniquely determined. It follows that A is a unique factorization domain (UFD), and in fact A is a principal ideal domain (PID). Indeed, every nonzero ideal of A is equal to

$$(\pi^n) = \{a \in A : v(a) \geq n\},$$

for some integer $n \geq 0$. Moreover, the ideal (π^n) depends only on n, not the choice of uniformizer π: if π' is any other uniformizer its unique representation $\pi' = u\pi^1$ differs from π only by a unit. The ideals of A are thus totally ordered, and the ideal

$$\mathfrak{m} = (\pi) = \{a \in A : v(a) > 0\}$$

is the unique maximal ideal of A (and also the only nonzero prime ideal of A).

Definition 1.12. A *local ring* is a commutative ring with a unique maximal ideal.

Definition 1.13. The *residue field* of a local ring A with maximal ideal $\mathfrak{m}$ is the field $A/\mathfrak{m}$.